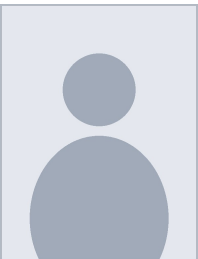


Nguyen Minh Quang

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🎓 scholar.google.com/citations?user=v25olzWAAAAJ



Research Experience

- Dec 2025 – Present
- Research Assistant

FPT University, Ha Noi, Vietnam — advised by Dr. Vanha Tran

 - Researched and applied Knowledge Graph and advanced Retrieval-Augmented Generation frameworks to evaluate corporate sustainability reports against the Global Reporting Initiative (GRI) and other ESG standards
 - Developed structure-aware chunking and LLM-guided information extraction pipelines; fine-tuned VLMs and domain-adapted embeddings model to accurately extract tabular and unstructured multi-modal data from complex ESG reports
 - Optimized KG retrieval and vector-space search pipelines, experimenting with heterogeneous data sources to enhance multi-hop reasoning, context grounding, and question-answering performance for ESG compliance
 - First-authored three peer-reviewed papers from this line of work (IUKM 2026, ICCSA 2026, AIS 2026); presented at AIS 2026 in San Antonio, USA

Professional Experience

- Feb 2026 – Present
- Agentic UP (AI-Native Enterprise)

Role

AI Engineer

Duration

Feb 2026 – Present (8 months), team of 6

Tech Stack

Claude Code SDK, OpenAI SDK, LangGraph, FastMCP, LiteLLM, Langfuse, DeepEval, vLLM, Qdrant, Neo4j

Responsibilities

 - Built a knowledge graph (~100K nodes, 400K+ edges; schema of 32 node types and 46 edge types) from 20+ internal systems — code, logs, and design documents — so agents can look up context instead of searching the codebase turn by turn
 - Built the retrieval layer on top of it: hybrid search (dense + sparse) in Qdrant with metadata filtering by source, entity type, and time, combined with graph traversal; reached Recall@10 of 0.93 and F1 of 0.90
 - Built a central gateway that gives every agent access to 300+ reusable skills across the software development lifecycle
 - Reduced prompt size by loading MCP tools on demand (tool search and CLI resolution across 10+ internal and SaaS servers), and by batching multi-step tool calls in a Python subprocess so only the summary reaches the model
 - Benchmarked models on our own agent tasks to pick the best cost–quality trade-off for each workload, and set prompt-cache lifetimes per session type (long for main sessions, short for sub-agents)
 - Built a session-analysis dashboard that reads agent traces and flags wasteful patterns — oversized context, cache misses, wrong model choice — with a suggested fix for each
 - Currently working on **self-improving agent memory**: turning execution traces and evaluation scores into updates to the knowledge graph and skills, without manual prompt changes

Impact

 - 65% of pull-request workflows automated
 - 10K–30K fewer tokens per session; 55–100% fewer tokens on multi-step tasks
 - 52% lower cost per session and 34% lower cost per 100 requests
- Jan 2024 – Jan 2026
- AI Engineer, GEM Corporation, Ha Noi, Vietnam

■ HSC AI Platform (HSC-AI)

Role

BE/AI Engineer (Core)

Duration

7 months, team of 7

Tech Stack

Flask, MySQL, LangChain, LiteLLM, AWS Bedrock, Azure OpenAI, Vertex AI, LangSmith, PyTorch, GitLab CI/CD

Responsibilities

 - Architected a multi-tenant SaaS AI platform (Flask, MySQL) using a shared-schema model with tenant-scoped data access; built core services for authentication, audit logging, and per-user history

Professional Experience (continued)

- Built a shared retrieval and orchestration layer (LangChain) powering agentic RAG, chatbots, real-time translation, and meeting summarization, so every new AI feature reuses one hardened pipeline
- Implemented a multi-provider LLM gateway on LiteLLM across Amazon Bedrock, Azure OpenAI, and Vertex AI with latency- and cost-aware routing and automatic fallback on provider errors and rate limits
- Instrumented end-to-end tracing and evaluation in LangSmith to monitor latency, token cost, and answer quality in production; shipped through GitLab CI/CD with automated test gates

Impact

- Serves 1M+ end users with p95 end-to-end latency held under the 5 s SLA
- Dynamic routing cut API latency by 40% and LLM spend by 70%; fallback kept the platform available through single-provider outages

■ *Enterprise Data Retrieval Chatbot*

Role AI Engineer **Duration** 6 months, team of 6

Tech Stack LangChain, Qdrant, Unsloth, vLLM, MLflow, Airflow, Flask, MongoDB, Redis

Responsibilities

- Built a hybrid retrieval system over enterprise documents with text and tabular images, routing each query between vector RAG (LangChain, Qdrant) and text-to-SQL with schema-aware prompting and read-only query validation
- Fine-tuned LLMs with Unsloth on Chain-of-Thought data for multi-step reasoning, benchmarked against the base model on a held-out evaluation set
- Designed a reproducible training-to-serving pipeline: Airflow for data and training orchestration, MLflow for experiment tracking and model registry, vLLM for streaming inference over WebSockets
- Used Redis for response caching and session state and MongoDB for conversation history to keep the serving path stateless and horizontally scalable

Impact

- One chatbot answers questions spanning unstructured documents, tabular images, and SQL databases
- Time-to-first-token under 8 s in production; 35% accuracy on multi-step queries over the base model

■ *Generative Perspective for Products*

Role BE/AI Engineer (Core) **Duration** 4 months, team of 8

Tech Stack FastAPI, PostgreSQL, Redis, PyTorch, SAM-3, Triton Inference Server, Milvus, Langfuse, Docker, AWS EC2

Responsibilities

- Researched and developed a TPS-based warp modeling algorithm for apparel images, optimizing per-product processing time from 3 s to 1.2 s
- Designed with BAs and POD clients a scalable backend (FastAPI, PostgreSQL) with Redis-backed asynchronous jobs for GPU-heavy segmentation and matting workloads
- Fine-tuned SAM-3 and developed a TPS-based warp modeling algorithm for apparel design automation
- Built a product recommendation service on SigLIP-2 embeddings and the Milvus vector database
- Served SAM-3, Sapiens, and ViTMatte on NVIDIA Triton with dynamic batching, containerized with Docker on AWS EC2 GPU instances, with request tracing in Langfuse

Impact

- ~98% segmentation accuracy ([metric, e.g. mIoU]) after fine-tuning
- Recommendations across a 1,000+ item catalog; p95 inference latency of 3 s in production

Nov 2023 –
Jan 2024

■ *AI Researcher Intern*, GEM Corporation, Ha Noi, Vietnam — advised by Dr. Tungxuan Vu

■ *Time-Series Forecasting of Mobile Data Usage*

Role AI Researcher Intern **Duration** Nov 2023 – Jan 2024, team of 2

Tech Stack Python, LSTM, XGBoost, Pandas, Scikit-learn

Responsibilities

- Built a reproducible data pipeline to clean telecom usage records and engineer demographic and historical-usage features
- Benchmarked LSTM and XGBoost under a common protocol with time-based train/validation splits to prevent look-ahead leakage

Impact

- Achieved R^2 of 0.946 on user-level data-usage prediction

Professional Experience (continued)

- Reduced RMSE by 25% versus baseline regression models

■ *Graph-Based Customer Segmentation from Call Networks*

Role AI Researcher Intern **Duration** Nov 2023 – Jan 2024, team of 2

Tech Stack Python, NetworkX, Louvain, Girvan–Newman, DBSCAN

Responsibilities

- Built a call-relationship graph from call-detail records enriched with demographic attributes
- Compared community-detection (Louvain, Girvan–Newman) and density-based (DBSCAN) methods, selecting partitions by modularity

Impact

- Identified 15 major customer clusters with distinct demographic and calling patterns
- Improved modularity by 15% over baseline; enabled the business team to target high-value and churn-risk segments

Education

2022 – 2025 ■ **B.S. in Information Technology**, Faculty of Artificial Intelligence
FPT University, Hoa Lac, Ha Noi, Vietnam GPA: 3.46/4.00

Publications

- 2026 ■ **M. Nguyen**, V. Tran. “News-Driven Sustainability Insight Extraction of Vietnamese Banking Using Knowledge Graph-RAG and LLM-Based Question-Answering System.” In *Proceedings of the International Symposium on Integrated Uncertainty in Knowledge Modelling and Decision Making (IUKM 2026)*, pp. 378–390. Springer Nature Singapore, Sep 2026. [link]
- **M. Nguyen**, A. Vuong, V. Tran, T. Bui. “GRI-Aligned Sustainability Reporting Assessment via Knowledge Graph-RAG and LLM-Based Question-Answering: A Case Study of Vietnamese Banks.” In *Proceedings of the International Conference on Computational Science and Its Applications (ICCSA 2026)*, pp. 100–116. Springer Nature Switzerland, Jun 2026. [link]
- **M. Nguyen**, V. Tran, T. Bui, T. Ngo, T. Nguyen. “Domain-Adapted Embedding Fine-tuning based ESG Question-Answering System Using GraphRAG and Large Language Models: A Case Study for the Vietnamese Banking Sector.” Presented at the *2nd International Conference on Artificial Intelligence Systems (AIS 2026)*, San Antonio, TX, USA, May 2026. Proceedings forthcoming.

Skills

Programming	■ Python, Rust, TypeScript, SQL, Bash
KG	■ LLM-guided entity/relation extraction, KG schema design, KG-enhanced RAG, Neo4j, Cypher, NetworkX
LLM & NLP	■ PyTorch, Hugging Face Transformers, LangChain / LlamaIndex, sentence-transformers, embedding fine-tuning, vector databases (Qdrant, Milvus), prompt engineering
ML Methods	■ Sequence models (LSTM), gradient boosting (XGBoost), community detection (Louvain, Girvan–Newman), density-based clustering (DBSCAN), feature engineering, model evaluation
MLOps & Data	■ NumPy, Pandas, Scikit-learn, Jupyter, MLflow, Airflow, Langfuse, DeepEval
Model Serving	■ ONNX, TensorRT, TorchScript, NVIDIA Triton Inference Server, vLLM
Cloud & Infra	■ AWS (S3, EC2, EKS, SageMaker), Docker, Git, Linux
Research	■ Experimental design, benchmark construction, ablation studies, literature review, academic writing

Honors & Certifications

- 2024 ■ **Top 3**, VPBank Hackathon — Talent Acquisition Assistant solution
- **Top 4**, FPT Hackathon — Generative AI solution
- IELTS Academic, overall band 6.5

References — contact details available upon request

Research advisor ■ **Dr. Vanha Tran**, Lecturer, FPT University — ORCID: 0000-0003-2714-6707

Internship supervisor ■ **Dr. Tungxuan Vu**, AI Specialist, GEM Corporation — Google Scholar